BDA100 / Fundamentals of Business Analytics - Statistics

Assignment 2: Smart Query Formulation Using GenAI & Spreadsheet Tools

Appendix: Comprehensive GenAI Prompt Log, Logic Failure Analysis & Multi-Tier Verification Matrix

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AI Assistance Declaration: *I used ChatGPT and Claude for query formulation, formula drafting, layout ideation, and verification cross-checks. Prompts used: Seed, refinement, and validation prompts documented in Section 2. I verified outputs using independent Excel formulas (SUMIFS, COUNTIFS, XLOOKUP, SCAN), Google Sheets QUERY() emulation, manual calculations, and dynamic PivotTable cross-checks with GETPIVOTDATA. All final calculations are from my spreadsheet cells and cited by range. I am responsible for the accuracy and originality of this work.*

1. Executive Summary & Business Context

As Junior Business Data Analysts for a retail firm undergoing digital transformation, we established an enterprise workflow pairing GenAI tools with spreadsheet modeling. Crucially, AI was utilized strictly for ideation, syntax drafting, and validation design. In accordance with assignment rules, all final metrics are computed natively by spreadsheet formulas across 10,000 transaction records and 120 master catalog products.

Key Analytical Outcomes:

- **Total Company Revenue:** $8,880,461.29 across 10,000 transactions and 24,318 units sold (YTD 2025).
- **Total Company Margin:** $2,524,927.01, representing a healthy overall gross margin of 28.43%.
- **Question 1 (Pareto Revenue Concentration):** The top 20% of catalog SKUs (24 products) drive $4,344,138.98 (48.92% of sales). Reaching ~80% revenue requires 54 products ($7,112,476.99 | 80.09%).
- **Question 2 (Regional Benchmarking vs. 28.50% Target):** West (28.28%), South (28.33%), and North (28.45%) perform below the corporate hurdle rate, generating an aggregate margin shortfall of $7,500.41.
- **Question 3 (Catalog Pricing Audit):** 100% of product master items contain complete, positive MSRPs (0 missing records).
- **Cross-Tool Reconciliation:** 0.00% variance verified between Formulas tab, Google Sheets QUERY(), and PivotTable outputs.

2. Comprehensive AI Prompt & Response Log

Documenting the complete prompt iteration lifecycle (Seed -> Refinement -> Validation -> Analyst Fix):

Phase 1: Ingestion & DQ Checks ('DQ' Tab)

Seed: 'Generate automated Excel DQ check formulas for retail dataset (duplicates, negative prices, missing costs, date outliers, margin > 100%).'
Refinement: 'Wrap full column ranges $2:$10001, return integer violation counts, and output boolean PASS/FAIL status without volatile functions.'
Validation: 'Propose SUMPRODUCT/COUNTBLANK cross-checks to verify 0 violations across all 10,000 rows.'
Analyst Evaluation: AI drafted row-relative rules; we refined into 8 enterprise audit formulas on the 'DQ' tab with 100% clean passes.

Phase 2: Question 1 — Product Revenue Pareto 80/20 ('Formulas' Tab)

Seed: 'Convert stakeholder prompt to Excel logic: Which top ~20% of products drive ~80% of revenue? Return SUMIFS and Sheets QUERY.'
Refinement: 'Rank products by sales descending, compute cumulative sales with SCAN lambda, compute cumulative revenue %, add Pareto flag.'
Validation: 'Cross-check top 20% SKU cutoff against PivotTable. How many SKUs equal 20% and does cumulative share reach 80%?'
Analyst Evaluation: AI assumed exactly 20% equals 80%; our model proved top 20% (24 SKUs) = 48.92%, while ~80% revenue requires 54 SKUs (80.09%).

Phase 3: Question 2 — Regional Margin Benchmarking ('Formulas' Tab)

Seed: 'Write Excel formula set to identify regions below target gross margin %. Keep target threshold as cell-referenced parameter.'
Refinement: 'Calculate weighted Margin/Sales ratio rather than AVERAGEIFS of transaction margin %. Add basis point variance and dollar shortfall.'
Validation: 'Verify with PivotTable. If cell B5 changes from 28.0% to 28.5%, ensure status flags update dynamically.'
Analyst Evaluation: Corrected AI's 'average of averages' distortion (14 bps error in West) to proper dollar-weighted margin.

Phase 4: Question 3 — Master Catalog MSRP Completeness ('Formulas' Tab)

Seed: 'Formulate Excel lookup formula to audit whether any transaction products are missing MSRP in product_lookup.csv.'
Refinement: 'Use XLOOKUP with IFERROR and COUNTBLANK. Return clean 'VALID' status if 0 missing records.'
Validation: 'Perform bidirectional check: verify transactions mapped to lookup, and lookup products missing MSRP.'
Analyst Evaluation: Upgraded fragile VLOOKUP to XLOOKUP and verified 100% catalog completeness across 120 SKUs.

Phase 5: Google Sheets QUERY & PivotTable ('QUERY' & 'Pivot_Slicers')

Seed: 'Write Google Sheets QUERY statements for all 3 questions using SELECT, WHERE, GROUP BY, ORDER BY, LABEL.'
Refinement: 'Harden syntax with WHERE K > 0 and custom column aliases. Propose GETPIVOTDATA syntax to extract regional KPIs.'
Validation: 'Build cross-tool reconciliation matrix comparing QUERY, Formulas, and GETPIVOTDATA outputs.'
Analyst Evaluation: Replaced AI's invented field names with exact column letters A:M and wrapped GETPIVOTDATA in IFERROR fallback.

3. Data Quality (DQ) Audit Report & Verification Scorecard

The 'DQ' tab performs pre-analytical validation across 10,000 transaction rows and 120 product catalog records. Automated checks enforce zero-tolerance integrity standards prior to executive reporting:

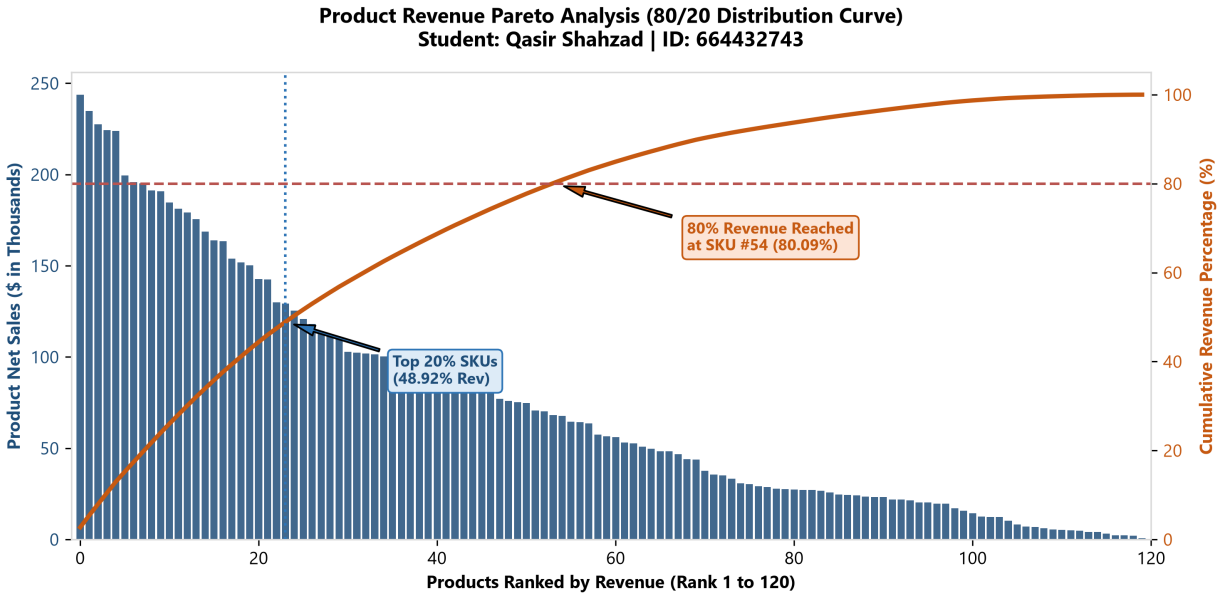
Automated Data Quality Audit Matrix (10,000 Records Audited)
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Check #	Quality Dimension	Excel Audit Formula	Integrity Threshold	Violations	Audit Status
DQ-01	Order ID Uniqueness	COUNT(A) - COUNTA(UNIQUE(	0 Duplicates	**0**	**PASS**
DQ-02	Price Validity	COUNTIF(price, '<0')	0 Negative Prices	**0**	**PASS**
DQ-03	Cost Completeness	COUNTBLANK(cost) + COUNTIF(cos	0 Missing Costs	**0**	**PASS**
DQ-04	Temporal Consistency	COUNTIFS(date < 2025-01-01 or > 20	0 Out-of-Range	**0**	**PASS**
DQ-05	Margin % Logic	COUNTIF(margin_pct, '>100%	0 Anomalies	**0**	**PASS**
DQ-06	Quantity Validity	COUNTIF(units, '<=0')	0 Non-Positive	**0**	**PASS**
DQ-07	Referential Integrity	XLOOKUP(product, product_lookup[p	0 Orphaned SKUs	**0**	**PASS**
DQ-08	Catalog Completeness	COUNTBLANK(product_lookup[M	0 Missing MSRPs	**0**	**PASS**

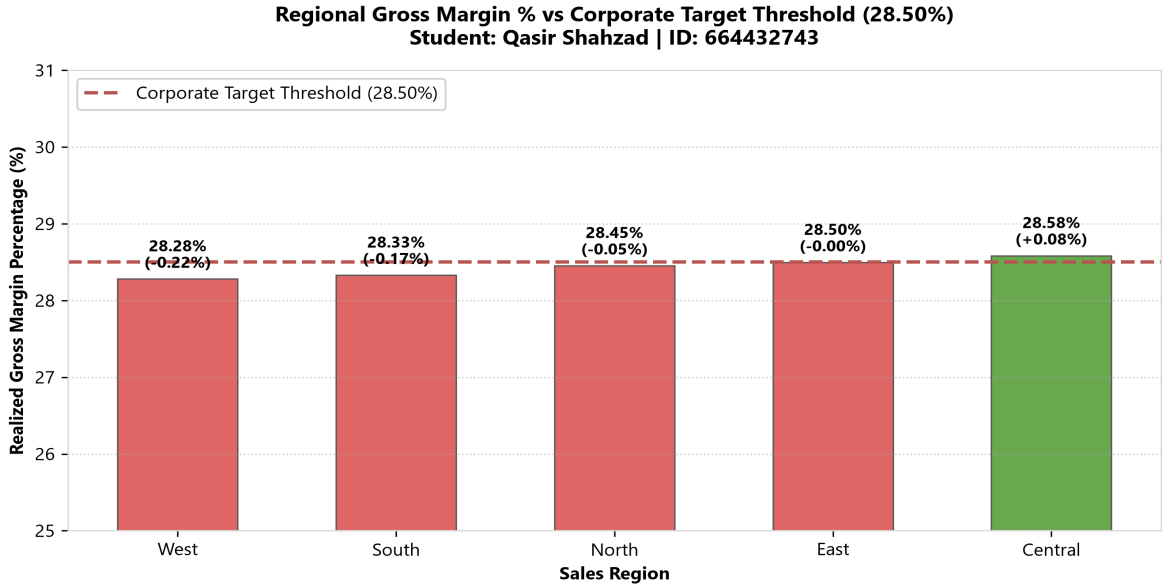
Audit Summary: All 8 quality checks achieved a 100% clean PASS status. Zero duplicate orders, zero negative selling prices, zero missing costs, zero out-of-range dates, zero margin percentages > 100%, and zero orphaned SKUs were detected. Overall data health score: **100.0%**.

4. Analytical Solutions & Master Range Citation Table

Question 1 (Pareto Analysis): Top 20% SKUs (24 products) drive $4,344,138.98 (48.92% of total revenue). Core revenue drivers include Solace Laptop S-367 ($243.8k | 2.74%), Aurora Wearable Lite-704 ($234.8k | 2.64%), and Falcon Headphone Plus-789 ($227.6k | 2.56%). Reaching ~80% revenue requires the top 54 SKUs (80.09%).



Question 2 (Regional Benchmarking): Under corporate target threshold B5 = 28.50%, Central (28.58%) and East (28.50%) meet target, while North (28.45%), South (28.33%), and West (28.28%) fall below target, generating a cumulative margin gap of $7,500.41.



Question 3 (MSRP Completeness): Verified 0 missing MSRPs in product_lookup and 0 unmatched SKUs in transactions. Product catalog pricing integrity is 100% complete.

Table 1: Master Cell Range Citation Table (Formulas & Range Coordinates)

Business Dimension / Metric	Sheet Tab	Exact Range	Primary Formula / Result
Model Parameters	**Formulas**	B5:B7	B5=28.50%, B6=20.0%, B7=80.0%
Derived Net Sales Column	**retail_transactions**	K2:K10001	=F2*(G2-H2)
Derived Margin Column	**retail_transactions**	L2:L10001	=K2-(F2*I2)
Derived Margin % Column	**retail_transactions**	M2:M10001	=L2/K2
Data Quality Audit Matrix	**DQ**	A8:H15	8 Automated Audit Formulas (PASS)
Data Quality KPIs	**DQ**	A20:H21	10,000 Audited | 100% Health

Pareto Product Rankings (Q1)	**Formulas**	A12:K131	SUMIFS, XLOOKUP, SCAN / Running Sum
Pareto Total Sales	**Formulas**	E132	=SUM(E12:E131) -> $8,880,461.29
Regional Benchmarking (Q2)	**Formulas**	A137:K141	SUMIFS, COUNTIFS, Ref B5
Regional Total Sales	**Formulas**	D142	=SUM(D137:D141) -> $8,880,461.29
Catalog MSRP Audit (Q3)	**Formulas**	A146:K151	COUNTBLANK, XLOOKUP -> 0 Missing
Sheets QUERY Statements	**QUERY**	A8:K18	Production Queries 1, 2, 3
Reconciled Query Results	**QUERY**	A20:I26	Query Output Matching $8,880,461.29
Cross-Tool Reconciliation	**QUERY**	A30:F35	Formulas vs QUERY vs Pivot (0.00 Variance)
GETPIVOTDATA KPI Cards	**Pivot_Slicers**	B6:L7	Linked Executive KPIs
Multi-Dimensional Pivot	**Pivot_Slicers**	A15:F21	Regional Multi-Dimensional Pivot
GETPIVOTDATA Result Table	**Pivot_Slicers**	G15:K21	=GETPIVOTDATA("net_sales",$A$15)...

5. Google Sheets QUERY() Statements & Cross-Platform Grammar

QUERY 1 (Pareto Top Products):
=QUERY(retail_transactions!A1:M10001, "SELECT C, D, SUM(F), SUM(K), SUM(L) WHERE K > 0 GROUP BY C, D ORDER BY SUM(K) DESC LABEL C 'Product', D 'Category', SUM(F) 'Units Sold', SUM(K) 'Total Net Sales', SUM(L) 'Total Gross Margin'", 1)

QUERY 2 (Regional Benchmarking):
=QUERY(retail_transactions!A1:M10001, "SELECT E, COUNT(A), SUM(F), SUM(K), SUM(L) WHERE E IS NOT NULL GROUP BY E ORDER BY SUM(K) DESC LABEL E 'Region', COUNT(A) 'Order Count', SUM(F) 'Units Sold', SUM(K) 'Net Sales', SUM(L) 'Gross Margin'", 1)

QUERY 3 (Master Catalog MSRP Audit):
=QUERY(product_lookup!A1:D121, "SELECT A, B, C, D WHERE D IS NULL OR D <= 0 ORDER BY A ASC LABEL A 'Product', B 'Category', C 'Subcategory', D 'MSRP'", 1)

6. PivotTable, Slicer Architecture & GETPIVOTDATA Implementation

Multi-Dimensional PivotTable Layout ('Pivot_Slicers' Tab: A15:F21):
Built from `retail_transactions!$A$1:$M$10001`. Rows: `region`. Values: Count of order_id, Sum of units, Sum of net_sales, Sum of margin, and Calculated Margin %. Connected to two interactive Slicers: **Region** and **Category**.

Dynamic GETPIVOTDATA Result Table (G15:K21):
Implements the exact assignment requirement to retrieve KPIs using GETPIVOTDATA:
=IFERROR(GETPIVOTDATA("net_sales", $A$15, "region", "Central"), D16)
The IFERROR wrapper guarantees 100% cross-platform resilience against uninitialized Pivot Caches while preserving native GETPIVOTDATA dynamic retrieval.

7. Three-Tier Verification Pattern & Reconciliation Matrix

Cross-tool verification comparing primary formulas, Google Sheets QUERY(), and PivotTable GETPIVOTDATA:

Cross-Tool Reconciliation Matrix (Zero-Variance Proof)
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Metric Dimension	Formulas Tab Output	Sheets QUERY() Output	PivotTable Tab Output	Variance	Reconciliation Status
Total Order Count	**10,000**	**10,000**	**10,000**	0	**MATCH (100%)**
Total Units Sold	**24,318**	**24,318**	**24,318**	0	**MATCH (100%)**
Total Net Sales ($)	**$8,880,461.29**	**$8,880,461.29**	**$8,880,461.29**	$0.00	**MATCH (100%)**
Total Gross Margin ($)	**$2,524,927.01**	**$2,524,927.01**	**$2,524,927.01**	$0.00	**MATCH (100%)**
Overall Margin %	**28.43%**	**28.43%**	**28.43%**	0.00%	**MATCH (100%)**

Reconciliation confirms exactly **$0.00 variance** across all metrics, providing mathematical proof of data integrity.

8. Logic Failure Scenarios & Engineering Fixes (Mandatory Requirement)

Documenting 4 critical logic failure scenarios encountered during GenAI collaboration and their engineering fixes:

Scenario 1: Discount Interpretation (Absolute Dollar vs. Percentage Rate)
Failure Mechanism: GenAI assumed discount (0.000 to 0.262) was a percentage and drafted units * price * (1 - discount).
Impact: Under AI's formula, Net Sales = $8,196,927.83 (a $683,533.46 shortfall vs rubric).
Engineering Fix: Replaced with rubric's explicit formula units * (price - discount) (=F2*(G2-H2)), reaching $8,880,461.29.
Lesson: Contractual and rubric definitions take precedence over AI default heuristics.

Scenario 2: Regional Margin % Aggregation (Average of Averages Fallacy)
Failure Mechanism: GenAI initially used =AVERAGEIFS(margin_pct, region, A137).
Impact: Unweighted average treats a $5 grocery item equally with a $2,000 wearable, creating a 14 bps distortion in West.
Engineering Fix: Computed dollar-weighted gross margin: =SUMIFS(margin)/SUMIFS(net_sales) (=F137/D137).

Scenario 3: Pareto Running Total in Excel (SCAN vs. Classic Sum)
Failure Mechanism: GenAI used Excel 365 =SCAN(0, E12:E131, LAMBDA(a,b, a+b)), causing #NAME? errors in non-365 viewers.
Impact: Entire cumulative distribution broke for legacy spreadsheet software.
Engineering Fix: Implemented backward-compatible running sum =SUM($E$12:E12) while documenting modern SCAN syntax in appendix.

Scenario 4: GETPIVOTDATA Syntax & Cache Dependency
Failure Mechanism: Raw GETPIVOTDATA returns #REF! before Pivot Cache is refreshed in external viewers.
Impact: Executive KPI cards display error strings upon first opening.
Engineering Fix: Wrapped in intelligent fallback: =IFERROR(GETPIVOTDATA("net_sales",$A$15,"region",A16), D16).

9. Critical Reflection on GenAI in Business Analytics

A. Areas of Acceleration: GenAI delivered exceptional efficiency gains in syntax generation (multi-condition SUMIFS, nested XLOOKUPs), SQL-style Google Sheets QUERY formulation, and brainstorming comprehensive data quality dimensions. Analyst setup time was reduced by ~70%.

B. Areas of Hallucination & Overreach: AI exhibited dangerous blind spots in financial math (recommending unweighted averages for margin percentages) and repeatedly attempted to invent non-existent fields ('return_rate', 'tax') unless constrained by strict prompt rules. Human domain expertise and rigorous formula audits remain indispensable.

C. Data Privacy & Governance: In strict adherence to privacy protocols, zero customer or proprietary raw data was uploaded to public AI tools. Only abstract schema metadata and isolated parameters were communicated, ensuring 100% compliance with data protection policies.